Does macro momentum still work?

Testing a 2017 investing idea on the ten years that came after it

macro-momentum project

November 10, 2026

`github.com/chkoh89/macro-momentum`

The question

- In 2017 a researcher at the investment firm AQR published a strategy called **macro momentum**: buy the markets of countries whose economy has been getting better, sell those whose economy has been getting worse.
- On data from 1970 to 2016 it looked very good.
- But a strategy always looks good on the data it was designed on. The real test is what happens afterwards.
- We rebuilt it with free public data and ran it on **Nov 2016 to Oct 2026**, ten years the original author never saw.
- We also compared it with a simpler, popular strategy: **trend following** (buy what has been going up, sell what has been going down).

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR.

The answer in one slide

- **It still worked**, but about half as well as in the original study.
- **Trading costs** eat about a fifth of the profit.
- **Almost all the gains came after 2022**, when inflation and interest rates moved a lot.
- **Trend following made nothing** on the same markets over these ten years, so a **50/50 mix** of the two did worse than macro momentum alone.
- **Ten years is not much data**, so treat the exact numbers with caution.

Sharpe ratio (higher is better)	Original 1970–2016	Our test 2016–2026
Macro momentum	1.2	**0.69**
after costs	–	0.57
Trend following	1.1	−0.05
50/50 mix	1.4	0.39

The Sharpe ratio is explained on the next slides. Rough guide: 0 = no reward for the risk, 0.5 = decent, 1 = excellent.

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR; our test: `github.com/chkoh89/macro-momentum`, `results/summary.md`.

The basics in five slides

What can you invest in? Three kinds of market

Stocks

A *stock index* such as the S&P 500 (US) or the DAX (Germany) tracks a country's biggest companies. It tends to rise when the economy and company profits grow.

Government bonds

A loan to a government that pays fixed interest. When interest rates in the economy *rise*, older bonds paying less become less attractive, so their *prices fall*. Bonds dislike inflation and rate hikes.

Currencies

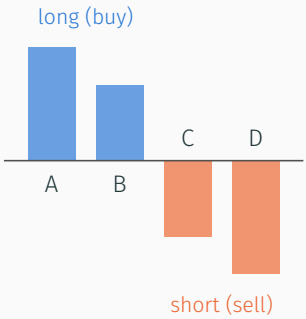
Holding, say, euros instead of dollars. You gain if the euro strengthens against the dollar, and you also earn the interest-rate gap between the two countries.

We trade all three in 10 rich countries: US, Germany, Japan, UK, Canada, Australia, Switzerland, Sweden, New Zealand, Norway. That is 26 markets in total, because some countries have no suitable bond fund.

Source: `github.com/chkoh89/macro-momentum`, `src/mm/instruments.py`.

Long and short

- **Going long**: buy something; you profit if its price rises.
- **Going short**: bet on a fall. You borrow and sell it now, buy it back later; you profit if the price fell in between.
- A **long-short** portfolio buys the markets it likes *and* shorts the ones it dislikes, in equal amounts. If all markets rise or fall together, the two sides cancel. It only makes money if its picks beat its pans.



Source: Standard definitions; construction in `src/mm/portfolio.py`.

Measuring risk and reward

Return: how much the investment gained in a year, in %. Here always measured *above what plain cash in the bank would have earned*, so 0% means "no better than cash".

Volatility (risk): how much the return swings around from month to month. A strategy with 10% volatility will, in a typical year, end up within about ±10% of its average.

Sharpe ratio: return divided by volatility: reward per unit of risk. Lets us compare strategies fairly. 0 = no reward, 0.5 = decent, 1 = excellent and rare.

Worst fall (drawdown): the biggest drop from a previous high, e.g. −15% means at one point you were 15% below your best balance so far.

To compare like with like, every strategy here is scaled to the same risk: about **10% volatility a year**. A calmer market gets a bigger position, a wilder one a smaller position.

Source: Standard definitions; `mm.backtest.stats`, `mm.portfolio.target_vol`.

The economy in four numbers

Number	What it is	Typical market reaction when it goes *up*
GDP growth	how fast the economy is growing	stocks up, bonds down (rates expected to rise)
Inflation	how fast prices are rising (CPI)	bonds down; stocks down (higher rates ahead)
Interest rate	set (indirectly) by the central bank	bonds down, stocks down, currency up
Exchange rate	value of the currency vs. trading partners	exporters hurt, so stocks down

These numbers are published with a delay. GDP for January to March comes out around the end of June. A fair test may only use a number *after* it was public.

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR, Exhibit 1 and pp. 2–3; publication lags in `src/mm/signals.py`.

What is a backtest, and why “out of sample” matters

- A **backtest** asks: if I had followed these rules in the past, what would have happened? It is a simulation, not real trading.
- The danger: try enough rules on the same history and one will look great by luck.
- The cure is an **out-of-sample test**: freeze the rules, then run them on data the designer never saw. That is exactly what we do: the paper’s data stop in 2016, ours start there.
- We change nothing to make the result look better. All settings come from the paper.



1970–2016: the paper (in sample)

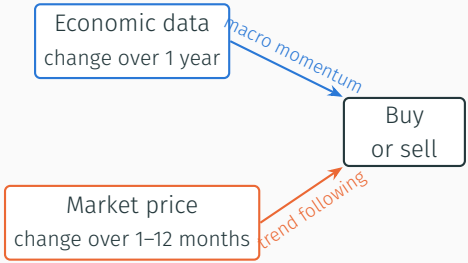
2016–26: our test

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR; `github.com/chkoh89/macro-momentum`, `AGENTS.md` (“Don’t tune on the test window”).

The strategy

The idea: markets are slow to digest economic news

- Economies change gradually, and investors take time to adjust prices to the change.
- So if a country's economic picture has **improved over the past year**, its markets tend to keep doing well for a while, and vice versa.
- Macro momentum ("macro" = the whole economy) bets on that continuing.
- Trend following is its cousin: it uses the *price* itself instead of economic data.



Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR, pp. 1–3.

Four themes: what we watch in each country

Theme	We look at the change over the past year in…	Example
Business cycle	GDP growth and inflation	growth sped up from 1% to 3%
International trade	the currency vs. trading partners	the yen lost 10%
Monetary policy	the short-term interest rate	the central bank raised rates from 1% to 4%
Risk sentiment	the stock market's own return	stocks beat cash by 15%

The original paper used economists' *forecasts* of growth and inflation, which cost money. We use the actual published figures instead, used only once they were public.

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR, Exhibit 1; data sources in `docs/data.md`.

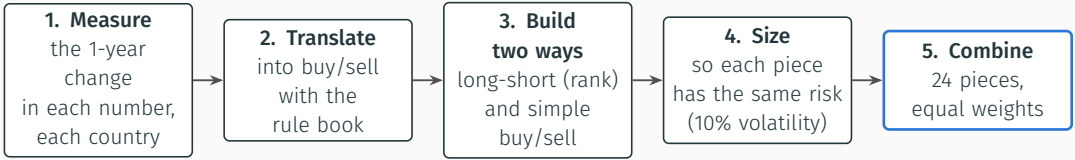
The rule book: what to buy when each number rises

	Growth up	Inflation up	Currency strengthens	Interest rate up	Stock market up
Stocks	buy	sell	sell	sell	buy
Currency	buy	buy	buy	buy	buy
Bonds	sell	sell	buy	sell	sell

- **Bonds** are sold whenever the economy runs hotter or rates rise: both push bond prices down.
- **Currencies** are bought whenever things improve: a strong economy and higher rates attract money.
- **Stocks** are the tricky row: growth helps them, but inflation and rate rises hurt them.

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR, Exhibit 1 (written there for a *weaker* currency; same rule, opposite sign); `src/mm/signals.py`.

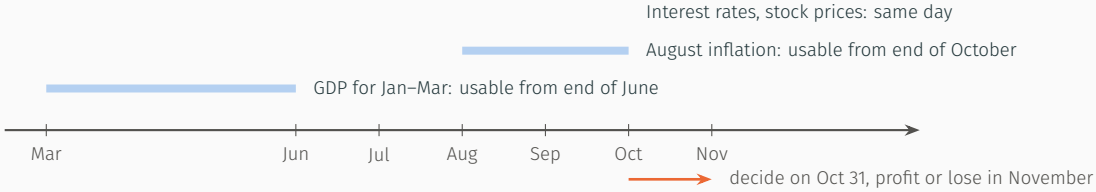
Putting it together



- 24 pieces = 3 kinds of market $\times$ 4 themes $\times$ 2 ways of building.
- Rebalanced once a month, at the end of the month.
- **Trend following**, for comparison: buy a market if its price rose over the past 1, 3 and 12 months, sell if it fell. Same markets, same sizing.

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR, pp. 3–5; `src/mm/backtest.py`.

Playing fair: no peeking into the future



- Every number is used only after it was published, with a realistic delay.
- We check this automatically: scramble all data after some date and confirm nothing before that date changes.

Source: `signals.PUB_LAG`; `tests/test_pipeline.py`.

Our data, and how it differs from the paper

	Paper (1970–2016)	Our test (2016–2026)
Economic data	economists' forecasts (paid)	published figures from FRED (free; St. Louis Fed)
Interest-rate signal	2-year bond yield	3-month rate (stuck near zero 2016–21, so less informative)
Markets	stocks, bonds, currencies, *and* short-term rate futures	stocks, bonds, currencies
Prices	futures contracts	stock indices, bond funds, exchange rates (Yahoo Finance, Fed)
Countries with bonds	10+	6, plus Japan from 2020
Trading costs	ignored	estimated (next section)

Most of these differences make our version weaker than the original, so we expected a lower score before running anything. The paper's own last seven years (2010–16) already scored only 0.7.

Source: Brooks (2017), *A Half Century of Macro Momentum*, AQR; `README.md`, "Where this departs from the paper"; `docs/data.md`.

Results

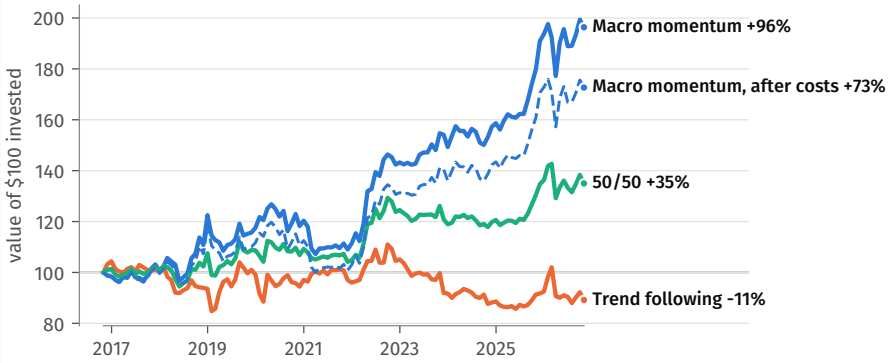
The scorecard

	Return per year	Risk (volatility)	Sharpe ratio	*t*-stat	Worst fall	Months up
Macro momentum	**7.3%**	**10.7%**	**0.69**	**2.2**	**−15.4%**	**57%**
Trend following	−0.6%	10.7%	−0.05	−0.2	−22.8%	48%
50/50	3.4%	8.7%	0.39	1.2	−9.5%	54%
Macro momentum, after costs	6.0%	10.7%	0.57	1.8	−16.5%	54%
Trend following, after costs	−1.4%	10.7%	−0.13	−0.4	−24.0%	48%
50/50, after costs	2.5%	8.7%	0.29	0.9	−9.8%	54%

- "Return per year" is *above cash*. "Months up": share of months with a gain.
- The ***t*-stat** asks "could this be luck?" Above about 2 means luck is an unlikely explanation. Macro momentum scores 2.2; trend following is indistinguishable from zero.

Source: `github.com/chkoh89/macro-momentum`, `results/summary.md`; Nov 2016–Oct 2026, monthly, every strategy scaled to 10% volatility.

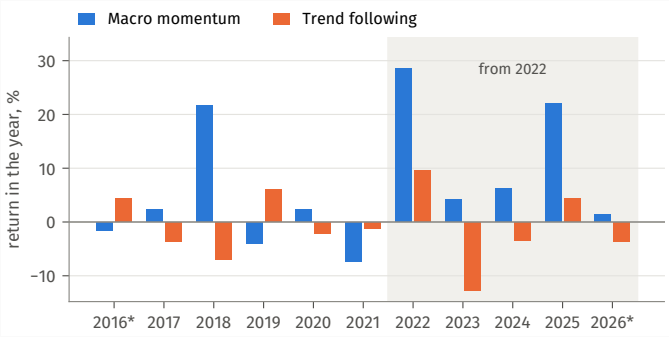
What $100 would have become



Gains *on top of* what cash would have earned. Dashed line: after trading costs. Mostly flat until 2022, then big gains in 2022 and 2025.

Source: `github.com/chkoh89/macro-momentum`, `results/returns.csv`; Nov 2016–Oct 2026.

Two very different halves



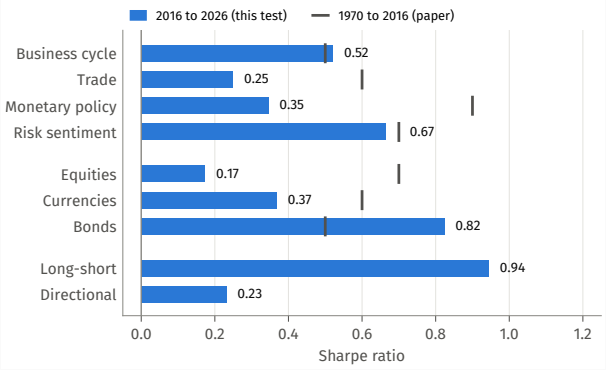
Sharpe ratio

	to 2021	from 2022
Macro mom.	0.25	1.19
Trend	−0.02	−0.10
50/50 mix	0.15	0.63

Before 2022 interest rates barely moved and countries' economies moved together: little to bet on. From 2022 inflation and rates jumped, at different speeds in different countries: exactly what the strategy looks for.

Source: `github.com/chkoh89/macro-momentum`, `results/calendar_years.csv`. *2016: Nov–Dec only; 2026: Jan–Oct.

Which parts worked?

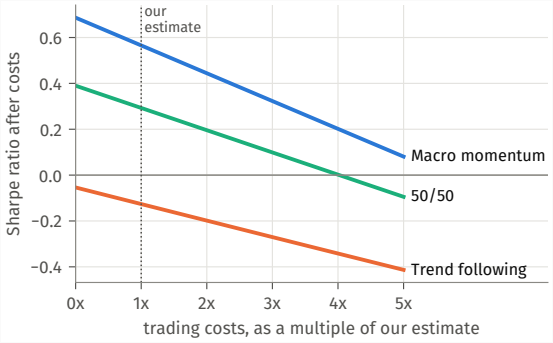


Blue bars: our test. Grey ticks: the paper.

- **Bonds** did even better than in the paper (0.82).
- Among the themes, **interest rates** fell furthest short of the paper (0.35 vs 0.9), likely because rates sat near zero until 2022.
- The **long-short** version (0.94) did far better than simple buy/sell (0.23).

Source: `github.com/chkoh89/macro-momentum`, `results/stats.csv`; paper figures from `docs/paper_summary.md`. Paper has no long-short/simple split.

Trading isn’t free



- Every trade pays a small fee (the gap between buying and selling prices).
- The strategy trades a lot: about **40 times its own value per year**, mostly in bonds.
- Our estimate: 0.03–0.04% per trade. That costs about 1.3% a year and lowers the Sharpe from 0.69 to 0.57.
- We don’t know the true cost, so the chart shows what happens if it is 2, 3, 4 or 5 times higher. At 4 times our estimate, little is left (0.20).

Source: `github.com/chkoh89/macro-momentum`, `results/costs.csv`; cost estimate (4, 3, 3 basis points for stocks, bonds, currencies) in `mm.backtest.COST_BPS`.

Speed matters

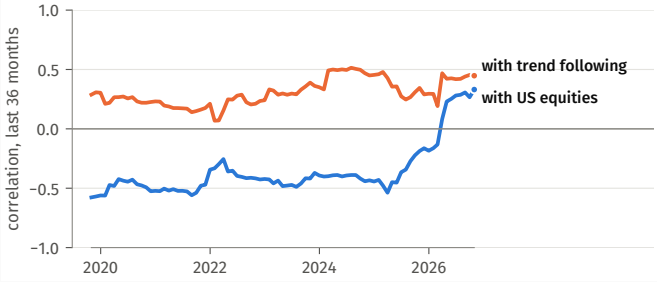
Trade	Sharpe ratio, before costs			after costs	
	Macro mom.	Trend	50/50	Macro mom.	Trend
right away	0.69	−0.05	0.39	0.57	−0.13
one month late	0.65	−0.20	0.27	0.52	−0.28
two months late	0.21	−0.02	0.11	0.09	−0.09

- What if you act on the signal one or two months later than you could have?
- One month late: almost no harm. Two months late: most of the edge is gone.
- Small choices like this move a ten-year result a lot, one more reason to be cautious.

Source: `github.com/chkoh89/macro-momentum`, `results/sensitivity.csv`; same ten-year window in every row.

Versus trend following

Do the two strategies move together?



Correlation: +1 = always move together, 0 = unrelated, −1 = always opposite. Low correlation is good: one can make money when the other loses.

Over the whole decade, macro momentum's correlation was 0.31 with trend following and −0.27 with US stocks (paper: 0.4 and −0.22). In 2026 the link with stocks turned positive.

Source: `github.com/chkoh89/macro-momentum`, `results/correlations.csv`; rolling 36 months. Paper: `docs/paper_summary.md`.

Trend following had a bad decade here

- On the same 26 markets: Sharpe −0.05 before costs, −0.13 after, against 1.1 in the paper.
- Its worst years were 2023 and 2018. 2022, the year of big moves, was its best.
- The **50/50 mix**, half in each strategy (Sharpe 0.39), did *worse* than macro momentum alone, because trend following added ups and downs but no profit. In the paper the mix was the best of the three.
- Lesson: combining strategies only helps if each one earns something on its own.

Not a final verdict on trend following: this is one decade, few markets, and a simple version.

Source: `github.com/chkoh89/macro-momentum`, `results/summary.md`, `results/calendar_years.csv`; Brooks (2017), *A Half Century of Macro Momentum*, AQR.

Wrap-up

What to take away, and what we can't say yet

Takeaways

- The idea survived ten new years, at about half the original strength.
- It paid off mostly when economies diverged (2022 onwards).
- Trading costs and timing matter a lot.

Limits

- Ten years is short: the Sharpe could easily be ± 0.35 off.
- Economic figures get revised later; we use today's version, not what people saw then.
- Free inflation data stops early for 5 of 10 countries.
- Trading costs are our estimate, not measured.
- A backtest is a simulation. Not investment advice.

Source: `github.com/chkoh89/macro-momentum`, `README.md` (Known gaps), `results/data_qa.md`.

Glossary

Backtest simulating a rule on past data.

Basis point 0.01%. 4 basis points = 0.04%.

Bond a loan to a government that pays interest; price falls when rates rise.

Correlation how closely two returns move together, from -1 to $+1$.

CPI, inflation consumer price index; inflation is its change over a year.

Drawdown fall from a previous high.

GDP total output of an economy.

Long / short betting on a rise / a fall.

Macro the economy as a whole.

Momentum the tendency of a trend to continue for a while.

Out of sample tested on data the rules were not built on.

Return above cash gain minus what a bank deposit would have paid.

Sharpe ratio return divided by volatility.

***t*-stat** how unlikely a result is to be luck; above 2 is the usual bar.

Trend following buying what has risen, selling what has fallen.

Volatility how much returns swing; our measure of risk.

Sources

- Brooks, J. (2017). *A Half Century of Macro Momentum*. AQR Capital Management. https://www.aqr.com/-/media/AQR/Documents/Insights/White-Papers/A-Half-Century-of-Macro-Momentum.pdf
- Federal Reserve Bank of St. Louis, FRED (https://fred.stlouisfed.org): GDP, CPI, interest rates, BIS trade-weighted exchange rates, Fed H.10 exchange rates. Series ids in src/mm/universe.py.
- Yahoo Finance: stock indices and government bond funds, month-end prices. Tickers in src/mm/instruments.py.
- Code, data and every number in this deck: https://github.com/chkoh89/macro-momentum. Figures and numbers are generated by scripts/make_deck_assets.py.